Negotiated Autonomy

Personal AI That Learns How We Delegate Spatial Work

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For discussion with Prof. Can Liu, City University of Hong Kong

The problem I want to study

I want to build personal AI that understands ongoing conversation, perceives the task context, contributes ideas and carries them into action. My research question is how a person and this increasingly capable collaborator negotiate what it is allowed to change. Understanding an intended outcome does not establish permission for every action that achieves it.

Consider a user redesigning a workspace: “Make it easier to collaborate, but keep the window workstation.” The AI improves the layout but also moves a monitor the user expected to remain untouched. Later, the user explicitly delegates rearranging the chairs while away. I want to understand both the unwanted expansion and the deliberate granting of greater autonomy.

My working proposition is that permission has geometry. In XR, a gesture marks the objects or region involved; a floating room model previews changes; an object-linked label states the intended operation and its duration. Spatial design makes delegation visible while retaining the broader ambition of a personal AI that works with someone across sessions.

Why this connects to your work

Reference-Based Manipulation [1] formalises spatial references through Source, Anchor and Frame. I want to use that structure to locate the spatial part of permission, then add the allowed operation and duration. EchoLadder [2] offers reversible scene suggestions; I want to study how users move between preview and delegated execution. Orality [3] captures thoughts as people develop them through speech; I want to identify how grants and revisions emerge within that speech.

Mixed-initiative interaction already treats autonomy as a negotiable design problem [4]. My proposed contribution concerns permissions on particular spatial objects, the interfaces through which people revise them, and how those agreements change with repeated use.

A prototype and three questions

The prototype will connect speech and editor state to scene-editing tools in a resettable desktop/XR room. People will choose among suggestions, ghost previews, sandbox edits and commitment to the shared scene. Background work will require an additional time-limited grant. An inspectable record will retain permission history rather than treating every preference as standing authorisation.

A user-specific estimator will learn preferred execution modes from confirmed corrections and approvals. It will operate inside explicitly granted permissions and propose extensions for confirmation. Silence will remain ambiguous evidence. A spoken stop or palm gesture will cancel queued work; manual takeover will pause changes to the touched object.

The first question is behavioural: how do people grant, retract and recalibrate delegation while working? The second is a design question: how should scope become visible, editable and interruptible? The longer-term question is how to distinguish wanted growth in delegation from autonomy drift as familiarity develops.

The first study I propose

In the first six months, I propose building two spatial permission interfaces and running a Wizard-of-Oz workspace study with 14 participants. Episodes involving an untouched object, a local correction and work continuing during absence will reveal how people communicate scope. Interaction records and event-based interviews will produce a taxonomy of authorisation behaviour and examples for the adaptive prototype.

The next study will compare fixed low autonomy, fixed high autonomy and adaptive boundaries with 30 participants; every condition will respect revocation and retain undo. Measures will cover task completion, autonomy drift, regret and repair effort. A six-week deployment with 12 participants will then trace how permissions change through use.

What I bring

My architectural training and CAVE installation experience connect spatial relationships with implemented interaction. I built Game Fellows, a bounded game companion, and prepared a Solo/Fellow comparison with paired random seeds, event logs and perceived-control questions; human evaluation is planned. My image-detection capstone also developed my practice of tracing failures to saved evidence. I would bring these experiences to a first study of how people delegate useful spatial work to personal AI.

Selected References

[1] Yangyang He, Zhuangze Hou, Yonglin Chen and Can Liu. 2026. Reference-Based Manipulation: A Framework and Pipeline for Multimodal Spatial Reasoning. UIST 2026 (accepted). doi:10.1145/3830398.3830657

[2] Zhuangze Hou, Jingze Tian, Nianlong Li, Farong Ren and Can Liu. 2025. EchoLadder: Progressive AI-Assisted Design of Immersive VR Scenes. UIST 2025. doi:10.1145/3746059.3747659

[3] Wengxi Li, Jingze Tian and Can Liu. 2026. Orality: A Semantic Canvas for Externalizing and Clarifying Thoughts with Speech. CHI 2026. doi:10.1145/3772318.3791713

[4] Eric Horvitz. 1999. Principles of Mixed-Initiative User Interfaces. CHI 1999, 159-166. doi:10.1145/302979.303030